higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 3

21 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1: COLD WATER RETICULATION

1.1
- The bore (inside diameter) and length of the pipe
- The roughness of the inside of the pipe
- The number of valves and fittings along the pipework
- The drop of water pressure due to loss of pressure in long runs of pipework (4)

1.2 When laying the potable water supply pipe next to a drain pipe, make sure that the pipes are at least 500 mm apart. If the pipes have to cross, the water supply pipe must be at least 100 mm above the drain pipe. (2)

1.3 It is the flow rate required in the pipework, measured in litres per second. (2)

1.4 1.4.1 Install thrust blocks.
1.4.2 Install anchor blocks.
(2 × 1) (2)

1.5 It is a detailed drawing, drawn to scale, of the residential cold water system that has to be installed.✓✓ It allows you to calculate and take off quantities or materials directly from the drawing.✓✓ (4)

1.6

WATER HAMMER	VIBRATIONS IN PIPEWORK
• Undersized piping • Incorrect valve installation • Loose piping • Trapped air in the pipes • Excessive drop in pressure in the system	• Excessive water velocity • Loose washer plate assembly • Faulty washer in tap, stopcock or mixer valve • Inlet pipe undersized.

(Any applicable causes – 2 × 2) (4)

1.7 The water should be:
- Safe to drink
- Attractive in appearance (sparkling)
- Free of taste
- Free of odour
- Usable in a variety of household and industrial purposes (Any 2 × 1) (2)

[20]

QUESTION 2: HOT WATER RETICULATION

2.1 2.1.1 Compression-type joint
2.1.2 GMS union

(2 × 1) (2)

2.2 When boiling water in a closed container changes to steam,✓ it may cause an immediate danger or safety hazard to the surrounding area.✓ If the water in a closed container overheats, the build-up of water pressure in the container could become too much and the container could explode.✓ This could cause much damage and perhaps also cause serious burns due to the hot steam in the immediate area around the container.✓ (4)

2.3
- The pressure control valve is separated into an inlet chamber and an outlet chamber by a washer that is held in position by a spindle and a pressure spring.
- This separation (closed position) is maintained due to the pressure of the water on the outlet side.
- When the hot water outlets are open, the pressure in the outlet chamber is reduced.
- The force of the spring then overcomes the force of the pressure in the outlet chamber and thus moves the valve off its seat to open the passage of water to the outlet chamber to the tap-off points.
- When the tap-off points are closed, the build-up of pressure on the diaphragm compresses the pressure spring and the washer seats into opening to the outlet chamber.
- This again isolates the inlet chamber from the outlet chamber and the pressure control valve. (Any 5 × 1) (5)

2.4 Thermal insulation on hot water pipes is a way of saving heat energy.✓ We cover hot water pipes✓ that are frequently used with a thermal insulation layer to keep most of the heat of the water inside the pipe.✓ When the pipe is used again, it is still warm✓ and it does not absorb as much of the heat of the water.✓ (5)

2.5 •A combination geyser uses low water pressure.
•A combination geyser has very little reserve water in the tank.
•A combination geyser cannot use modern mixers because of low pressure.
•Airlocks in hot water pipes are common in a combination geyser. (4)

[20]

QUESTION 3: ABOVE-GROUND DRAINAGE SYSTEMS

3.1
- They must not corrode (rust) or wear away.
- They must be unaffected by coastal air and industrial air pollution.
- They must be durable and not easily damaged by chemicals, industrial pollution and household cleaning products. (3)

3.2 Pipe clips✓ and pipe brackets✓ (2)

3.3.
3.3.1 Water closet
3.3.2 Bidet
3.3.3 Shower
3.3.4 Urinal
3.3.5 Wash hand basin
(5 × 1) (1)

3.4 A single-stack system is a one-pipe system in which the individual water traps for each soil and waste fixture are not vented individually.✓ All the soil and waste fixtures discharge directly into one discharge stack.✓ This stack serves as both the discharge pipe and the vent pipe for both soil water and waste water fixtures of the installation.✓ (3)

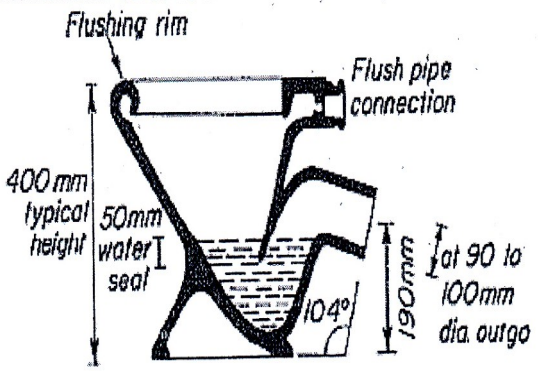
3.5
- It is very expensive to install this system.
- It requires more materials/fittings. (2)

[15]

QUESTION 4: SANITARY INSTALLATIONS

4.1 When water from a sink splashes against it, the water will splash back or run back into the sink.✓ This makes it easier to clean than a wall.✓ Splash backs are usually made of pressed stainless steel,✓ but it could be manufactured from any other suitable material,✓ for example, epoxy-coated fibreglass, marble, or even granite. (Any 4 × 1) (4)

4.2



Correctness	4
Label (any)	2

Wash-down WC pan: P-trap (6)

4.3 We fit a bath P-trap with a giraffe-neck overflow pipe to the outlet of the bath overflow. (3)

4.4 We usually use a junior flush valve to connect the water supply to a wall-mounted, single-bowl urinal. (2)
[15]

QUESTION 5: RAINWATER SYSTEM

5.1
- uPVC (unplasticised polyvinyl chloride)
- Fibre cement
- GMS (galvanised mild steel)
- Aluminium (Any 3 × 1) (3)

5.2 By using:
- Solder and a pop-rivet joint
- Gutter sealant and a pop-rivet joint
- An expansion joint or slip joint (3)

5.3. 5.3.1 Prevents the rainwater in the gutter from flowing down at the end of the gutter

5.3.2 Is commonly used on gutters when one roof hangs over another roof

5.3.3 Is used to discharge and direct the water away from the external wall of a building

(2 × 3) (6)

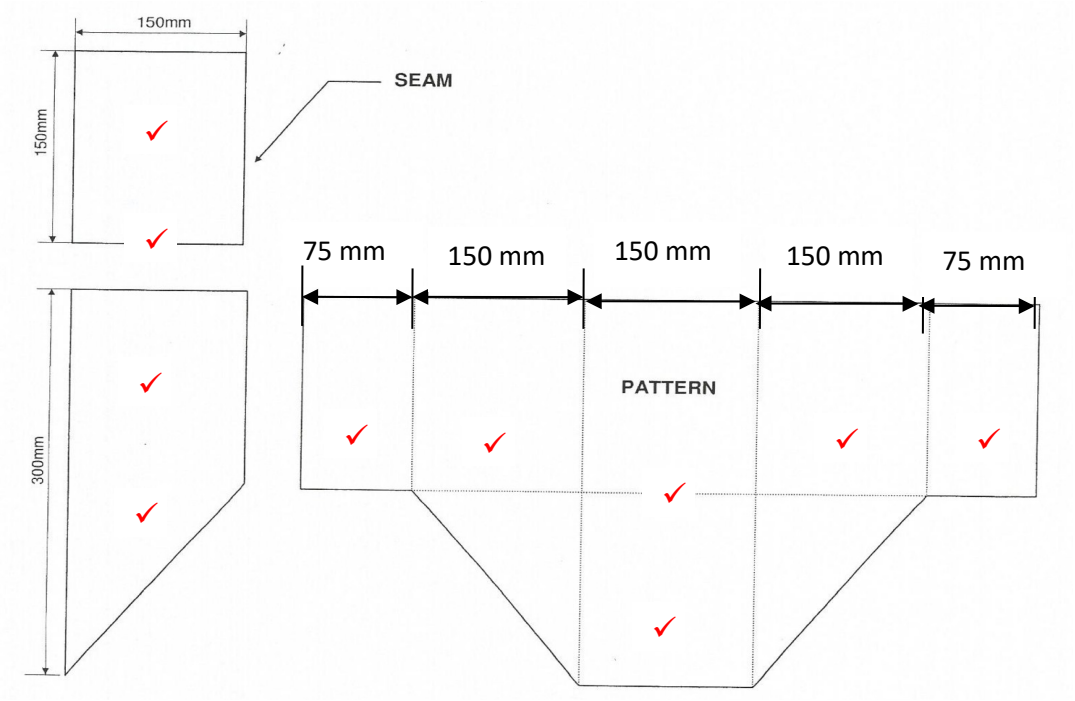
5.4 The water discharges into a storm-water drain. (1)

5.5 5.5.1 To prevent bad smell or foul air to enter the house/building

5.5.2 To catch or prevent fatty water or grease water to enter the drain, causing blockage

(1 × 2) (2)
[15]

QUESTION 6: VENT-PIPE FLUSHING



[10]

QUESTION 7: TRENCHING, EXCAVATION AND TRENCH SUPPORT

7.1 Setting out a trench to line and level means to prepare the drainage layout,✓ the depth of the drain,✓ the gradient✓ and the dimensions of the drain according to the drawings.✓ (4)

7.2 For a 100-mm-diameter drain pipe, it is 1 : 40. (1)
[5]

TOTAL: 100

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